YILDIZ TECHNICAL UNIVERSITY

FACULTY OF ELECTRICAL AND ELECTRONICS

ELECTRICAL ENGINEERING DEPARTMENT



ELM3091 DIGITAL SYSTEMS COURSE PROJECT REPORT

Project General Information	
Subject:	**Inverse Traffic Light Controller**
Team Members:	**1. [redacted] – Samet Erhan Sarı** **2. [redacted] – Burak Kürklü** **3. [redacted] – Oğuz Esat Kurt**
Deadline:	**09.06.2026**

1. Literature Overview

Traffic light controllers represent one of the most fundamental pedagogical and practical applications in digital electronics literature, primarily utilized to validate the theoretical principles of Synchronous Sequential Logic Circuits and Finite State Machines (FSM). Although modern industrial traffic management systems are predominantly driven by software running on microprocessors, microcontrollers (such as the Arduino platform), or Programmable Logic Controllers (PLCs), hardware-only state machines built from discrete Transistor-Transistor Logic (TTL) integrated circuits and bistable memory elements remain the architectural core of digital synthesis.

In digital design, state machines are classified into two major architectural models: Mealy and Moore. In a Mealy machine, outputs are a function of both the current state and the current inputs, which can occasionally introduce unwanted transient glitches or noise propagation into the output stage. Conversely, in a Moore machine, the outputs depend strictly on the current internal state of the system. For hardware-implemented synchronous designs, the Moore architecture is highly preferred because it provides a remarkably stable and clean output waveform, completely isolated from direct input fluctuations. The traffic light controller developed in this project inherently follows the classical Moore Finite State Machine model, as the output states of the signaling LEDs are determined solely by the stored state bits passed through combinational decoding logic.

Selecting the appropriate memory element is a critical design decision in the hardware realization of an FSM. Among the bistable devices widely analyzed in literatüre; namely SR, JK, T and D-type flip-flops; the D (Data/Delay) flip-flop stands out due to its characteristic behavior of transferring the input data directly to the output at the triggering clock edge. This feature significantly simplifies the derivation of next-state excitation equations. The standard academic workflow for synthesizing such an FSM involves a rigorous sequence of engineering steps:

1. Constructing a comprehensive State Transition Diagram that maps out the operational phases of the system.
2. Assigning unique binary state codes to each traffic phase.
3. Deriving a State Transition Table that mathematically links the present states to the next states.
4. Implementing Karnaugh Maps (K-Maps) or Boolean algebraic simplification to minimize the excitation equations, thereby reducing the physical logic gate count.
5. Physically wiring the combinational logic gates and memory components on a breadboard based on the minimized expressions.

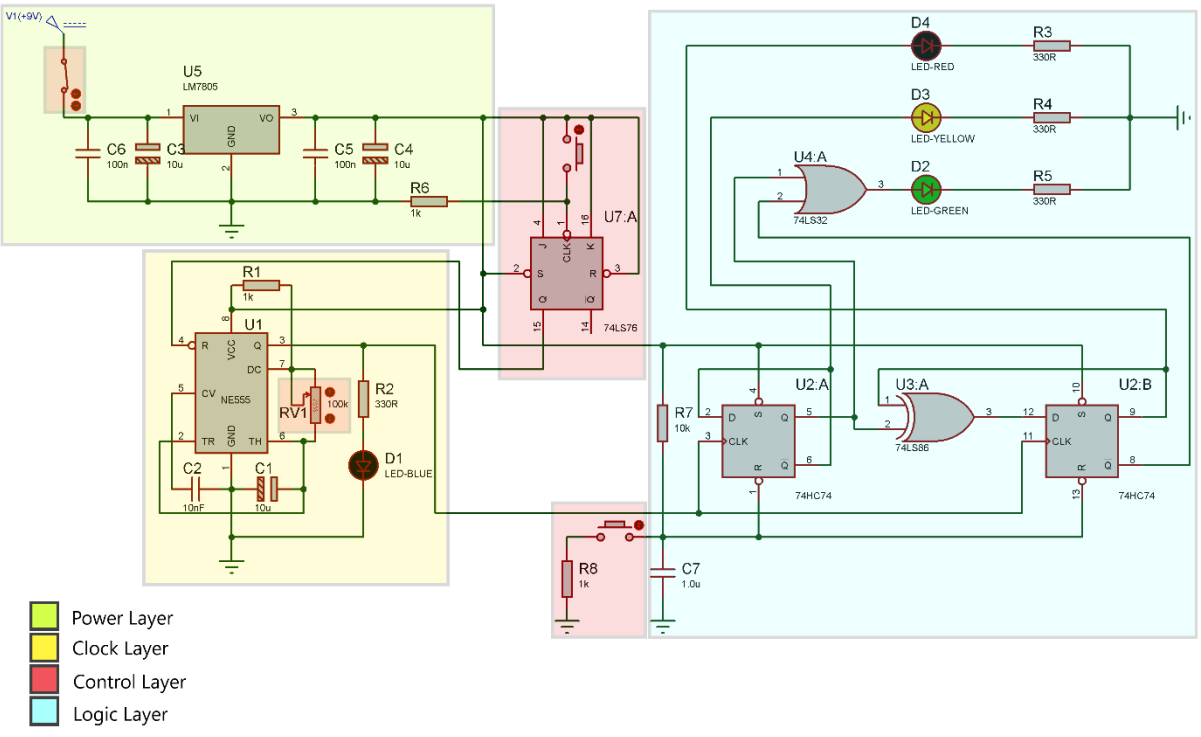
To mitigate severe physical hardware constraints known as "Clock Skew" and cumulative "Propagation Delay", sequential circuits must be designed with a strictly synchronous topology. In asynchronous (ripple) counters, each flip-flop is clocked by the output of the preceding stage, causing a cascading delay that grows linearly with the number of bits, often

resulting in momentary erroneous intermediate states (transient hazards). In a synchronous configuration, however, the clock (CLK) inputs of all memory elements are tied together to a single, common clock distribution line. Consequently, all state transitions occur simultaneously on the exact same clock edge, unifying the propagation delays and ensuring that state changes settle in a stable manner within a fraction of a microsecond.

When engineering discrete hardware systems, mechanical switch bouncing (or chatter) poses a significant real-world challenge. When a mechanical button is pressed, the elasticity of the internal metallic contacts causes thousands of microscopic electrical arcs and fluctuations for a brief period. Digital ICs process each of these rapid fluctuations as a separate logical transition or clock pulse, causing the state machine to erratically jump into random states. In professional engineering, this issue is systematically addressed under the domain of "Signal Conditioning" using hardware-based RC debouncing filters or Schmitt-trigger buffers. The integration of power stabilization, precision clock oscillation, signal conditioning, and synchronous counter logic in this project demonstrates a comprehensive synthesis of academic digital design theory and practical engineering implementation.

2. Project Circuit Scheme and Explanations

The physical hardware architecture implemented in this project consists of four functional layers meticulously designed to govern electrical stability, timing accuracy, user control, and state calculation:



2.1. Power Layer and Linear Voltage Regulation

The primary energy source for the system is a nominal 9V DC battery. However, the TTL and CMOS digital logic integrated circuits utilized across the design (74LS and 74HC families) require a highly stable supply voltage of $5\text{V} \pm 0.25\text{V}$to prevent dielectric breakdown of the gate oxide layers and to ensure predictable logic threshold levels (V_{IH}, V_{IL}). This precise electrical conversion is achieved using the LM7805 three-terminal linear voltage regulator. The LM7805 drops the excess input voltage by dissipating it as heat across its internal pass-transistor structure, outputting a continuous, regulated 5V DC rail.

To isolate the regulation process from high-frequency electromagnetic interference (EMI) and transient load variations, specialized filtering capacitors are integrated into the power rail topology:

- Two 100nF Ceramic Capacitors (Parallel to Input/Output): Featuring an exceptionally low Equivalent Series Resistance (ESR), these components act as high-frequency bypass elements, shunting high-frequency noise and line ripples directly to the ground plane.
- Two 10µF Electrolytic Capacitors (Parallel to Input/Output): These components function as local bulk energy reservoirs. When the logic gates alter their states simultaneously (especially during the concurrent turn-on transitions of the output LEDs), they draw sharp, instantaneous current pulses from the supply. This transient current demand can cause local voltage drops due to trace impedance. The 10µF capacitors instantly release their stored charge into the rail during these sub-microsecond intervals, smoothing out voltage sags and preventing the digital ICs from experiencing spurious resets or entering forbidden logic states.

2.2. Control Layer and Master Reset Topology

The control layer is engineered to manage manual user interactions and guarantee a safe initialization sequence upon power-up. To prevent the memory states of the flip-flops from entering an unpredictable, undefined configuration when the circuit is first energized, a dedicated Master Reset Button is implemented.

This manual reset switch is integrated into a hardware-based RC charging network forming a specialized Power-On Reset and debouncing circuit. Based on transient circuit analysis, the time constant (τ) of this network is defined as:

$$\tau = R \cdot C = 10\ \text{k}\Omega \cdot 1\ \mu\text{F} = 10\ \text{ms}$$

In hardware design literature, a capacitor requires approximately 5τ to reach full charge and achieve a stable logic-1 state. Therefore:

$$5\tau = 5 \cdot 10\ \text{ms} = 50\ \text{ms}$$

When the circuit is initially switched on, the voltage rises from 0V to 5V over this 50ms interval. During this window, the capacitor is in a low-charge state, effectively presenting a logical Low (0) to the asynchronous clear (CLR) inputs of the 74HC74 flip-flops. Because the CLR input of the 74HC74 is active-low, it completely overrides the synchronous clock line, forcing the outputs to a clean zero state ($Q_1 = 0, Q_2 = 0$). This configuration entirely suppresses random startup states and ensures the FSM reliably commences from the 00 phase. When the capacitor finishes charging past the logic-1 threshold, the reset condition is lifted, and the counter begins normal synchronous operation. Pressing the button manually at any point instantly discharges the capacitor, forcing the CLR lines Low and restarting the sequence from the initial state.

Additionally, a 74LS76AN JK Flip-Flop is positioned at the front-end of the control layer to act as a debounced clock-enable latch. The J and K inputs are tied directly to the 5V rail (Logic 1-1), placing the device permanently in Toggle Mode. When its clock pin is pulsed via a mechanical push-button, the output switches state deterministically; this stable output is fed directly into the Reset pin (Pin 4) of the NE555 timer to act as a master start/stop mechanism for the clock generator.

2.3. Timer Layer (NE555 Oscillator Architecture and Frequency Characteristics)

The periodic timing pulses driving the state transitions are generated by an NE555 Timer IC configured as an Astable Multivibrator. The network utilizes a fixed resistor $R_1 = 1$ kΩ, a variable potentiometer $R_2 = 100$ kΩ for manual frequency tuning, and a timing capacitor $C_1 = 10\ \mu$F.

The internal structure and pin distribution of the NE555 operate as follows:

- VCC and GND (Pin 8 - Pin 1): Connected directly to the regulated 5V rail and system ground, biasing the internal voltage divider string which consists of three highly matched 5 kΩ resistors. This internal network establishes the rigid internal reference thresholds at $\frac{1}{3}V_{CC}$ and $\frac{2}{3}V_{CC}$.
- Control Voltage (Pin 5): Bypassed to ground through a 10nF ceramic capacitor to safeguard the internal comparator references from electrical noise.
- Reset (Pin 4): Connected to the Qoutput of the 74LS76AN latch; driving this pin Low immediately disables the internal discharge transistor and stalls oscillation.
- Trigger and Threshold (Pin 2 - Pin 6): Shunted together to continuously monitor the exponential voltage profile across the timing capacitor C_1. When the capacitor voltage charges up to $\frac{2}{3}V_{CC}$, the internal threshold comparator resets the internal SR latch. Conversely, when the voltage discharges down to $\frac{1}{3}V_{CC}$, the trigger comparator sets the latch, initiating a new cycle.

- Discharge (Pin 7): Connected to the junction of $R_1$and R_2, providing a controlled current path to ground during the capacitor's discharge phase.

By manually adjusting the R_2 potentiometer, the operating frequency (f) of the clock output sweeps across distinct empirical and theoretical benchmarks:

- At full 100kΩ resistance: The circuit oscillates at a slow rate of approximately 0.72 Hz.
- At 75kΩ resistance: The timing matches the standard 1 Hz mark (one clock pulse per second).
- At 50kΩ resistance: The frequency rises systematically to 1.42 Hz.
- At 25kΩ resistance: The transition speed accelerates to 2.82 Hz.
- At 0Ω (Potentiometer dialed to its lower limit): The frequency reaches its upper theoretical boundary of 144 Hz.

The mathematical expression governing the oscillation frequency (f) is defined as:

$$f = \frac{1.44}{(R_1 + 2R_2) \cdot C_1}$$

Consequently, the duty cycle (the ratio of high time to total period) is expressed as:

$$\text{Duty Cycle} = \frac{R_1 + R_2}{R_1 + 2R_2} \times 100\%$$

As the potentiometer resistance R_2 approaches zero to increase the frequency, the mathematical weight of the $2R_2$term in the denominator diminishes rapidly, causing the duty cycle to converge toward 100%. Ideally, at $R_2 = 0\Omega$, the output would settle into a constant DC High state. However, due to the inherent residual internal resistance present in physical potentiometer hardware, the discharge path (Pin 7) never drops to an absolute zero ohm. Consequently, the NE555 continues to oscillate at a rapid 144 Hz. Because the resulting Low interval is incredibly brief, the off-time of the blue indicator LED (D1) falls far below the response time of the human eye. Due to the phenomenon of persistence of vision, the human eye blends these rapid pulses, making the blue LED appear continuously illuminated.

The resulting synchronous square-wave clock from Pin 3 is distributed in parallel to the clock inputs of the sequential logic layer.

2.4. Logic Layer (Counter Synthesis and Edge-Triggering Reliability)

The logic layer contains the core processing hardware, combining a 2-bit synchronous state counter and a combinational decoding matrix. It comprises two 74HC74 D-type Flip-Flops, a 74LS86 XOR gate, and a 74LS32 OR gate.

A key engineering choice in this design is utilizing the 74HC74 (High-Speed CMOS) family instead of the standard legacy 74LS series for the primary memory cells. When the clock generator is tuned to its high-frequency limit (144 Hz), the rise and fall times of the clock edges become exceptionally steep and narrow. Standard LS series devices, which suffer from minority-carrier storage delays within their internal bipolar junction transistors, can fail to resolve such high-speed edges reliably, leading to missed counts or state corruption. The 74HC series, with its high-speed CMOS architecture and immense input impedance, easily captures these sub-nanosecond transitions, ensuring flawless state propagation across all frequency ranges.

The state bits are assigned such that the rightmost flip-flop represents the Most Significant Bit (MSB, Q_1) and the leftmost flip-flop represents the Least Significant Bit (LSB,Q_2). The next-state feedback loops are implemented as follows:

- 2. Flip-Flop (LSB - Left): The data input ($2D$) is cross-coupled directly to its own inverted output ($\overline{2Q}$), meaning $2D = \overline{2Q}$. This configuration creates a deterministic toggle behavior on every rising clock edge.
- 1. Flip-Flop (MSB - Right): The data input ($1D$) is driven by the output of the 74LS86 XOR gate, which processes the current true outputs of both stages ($1D = 1Q \oplus 2Q$).

This structural layout forces the counter to execute a flawless, repeating state sequence on successive clock edges:

$$\text{State Sequence:} \quad 00 \longrightarrow 01 \longrightarrow 10 \longrightarrow 11 \longrightarrow 00$$

2.5. State Analysis and Logical LED Scenarios

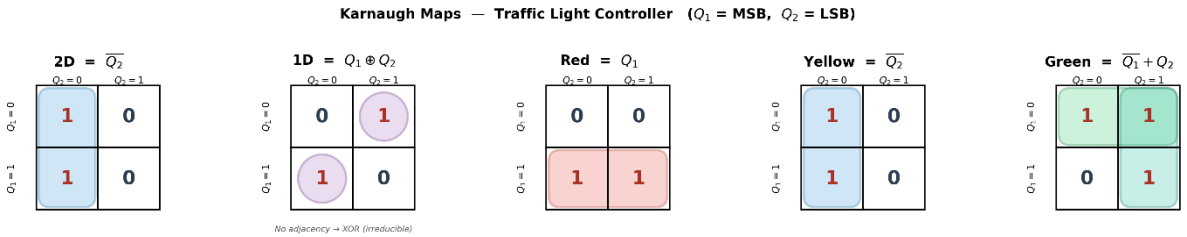
The outputs of the state counter (Q_1 andQ_2) are mapped into the combinational logic gates to drive the output LEDs according to the target traffic signaling protocol. The physical output equations are derived as follows:

- Red LED (D4): Tied directly to the Q_1 output of the MSB flip-flop (Red $= Q_1$).
- Yellow LED (D3): Fed directly by the inverted output ($\overline{Q_2}$) of the LSB flip-flop (Yellow $= \overline{Q_2}$).
- Green LED (D2): Driven by a 74LS32 OR gate that processes the true LSB output (Q_2) and the inverted MSB output ($\overline{Q_1}$), yielding (Green $= Q_2 + \overline{Q_1}$).

The following truth table demonstrates the logical operations and Boolean proofs for each operational phase:

Phase	Q1 (MSB)	Q2 (LSB)	Red Output (Q1)	Yellow Output (Q2)	Green Output (Q2+Q1)	Active LED Combination
State 00	0	0	0 (OFF)	1 (ON)	$0 + 1 =$ 1 (ON)	Yellow and Green
State 01	0	1	0 (OFF)	0 (OFF)	$1 + 1 =$ 1(ON)	Green Only
State 10	1	0	1 (ON)	1 (ON)	$0 + 0 =$ 0 (OFF)	Red and Yellow
State 11	1	1	1 (ON)	0 (OFF)	$1 + 0 =$ 1(ON)	Red and Green

The K-Maps For the Circuit:



Phase-by-Phase Boolean Verification:

- State 00 ($Q_1 = 0, Q_2 = 0$): The circuit is either initializing or has been reset. The Red LED is un-biased since $Q_1 = 0$. The Yellow LED receives $\overline{Q_2} = 1$, switching it instantly into conduction. The Green LED's OR gate sees $Q_2 = 0$ on its first input and $\overline{Q_1} = 1$ on its second input; evaluating the Boolean OR expression (0 OR $1 = 1$) drives the gate output High. Therefore, in this initial phase, the Yellow and Green LEDs are illuminated simultaneously.
- State 01 ($Q_1 = 0, Q_2 = 1$): At the first rising clock edge, Q_2 toggles to 1 while Q_1 remains 0. The Red LED remains un-biased ($Q_1 = 0$). The Yellow LED input drops to $\overline{Q_2} = 0$, turning it off. The Green LED's OR gate processes $Q_2 = 1$ and $\overline{Q_1} = 1$; since both conditions evaluate to High (1 OR $1 = 1$), the gate continues to supply current to the diode. Consequently, only the Green LED is illuminated.

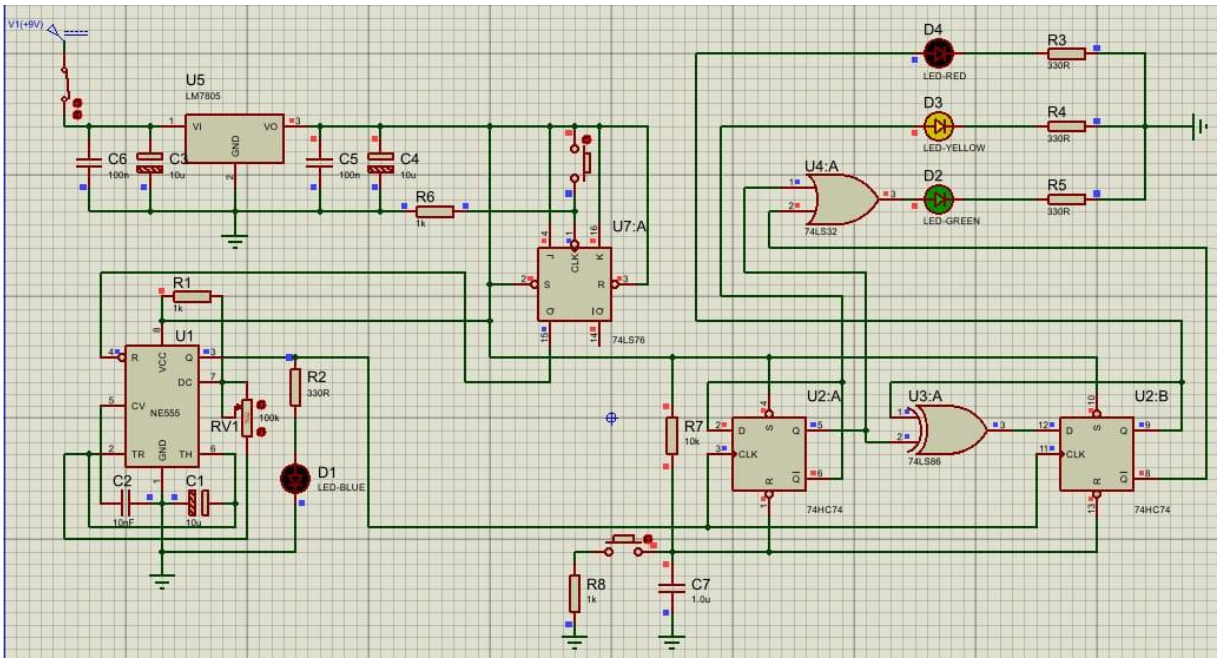
- State 10 ($Q_1 = 1, Q_2 = 0$): Upon the second clock pulse, the counter advances to the next binary stage. The Red LED line rises to $Q_1 = 1$, driving the red indicator into full conduction. The Yellow LED receives $\overline{Q_2} = 1$ due to the zero state of the LSB, turning it on. The Green LED's OR gate receives $Q_2 = 0$ and $\overline{Q_1} = 0$; because both inputs are at a logic low level (0 OR $0 = 0$), the OR gate output drops to zero, and the Green LED turns off. Thus, the Red and Yellow LEDs are illuminated simultaneously.
- State 11 ($Q_1 = 1, Q_2 = 1$): At the third clock pulse, the counter reaches its peak configuration. The Red LED remains active since $Q_1 = 1$. The Yellow LED turns off because its input drops to $\overline{Q_2} = 0$. The Green LED's OR gate evaluates $Q_2 = 1$ and $\overline{Q_1} = 0$; because the first input is High (1 OR $0 = 1$), the gate switches back to a high output state, re-energizing the green channel. In this final phase, the Red and Green LEDs are illuminated simultaneously. When the fourth clock pulse arrives, the logic loops back seamlessly to State 00, repeating the cycle indefinitely.

3.MATERIAL LIST

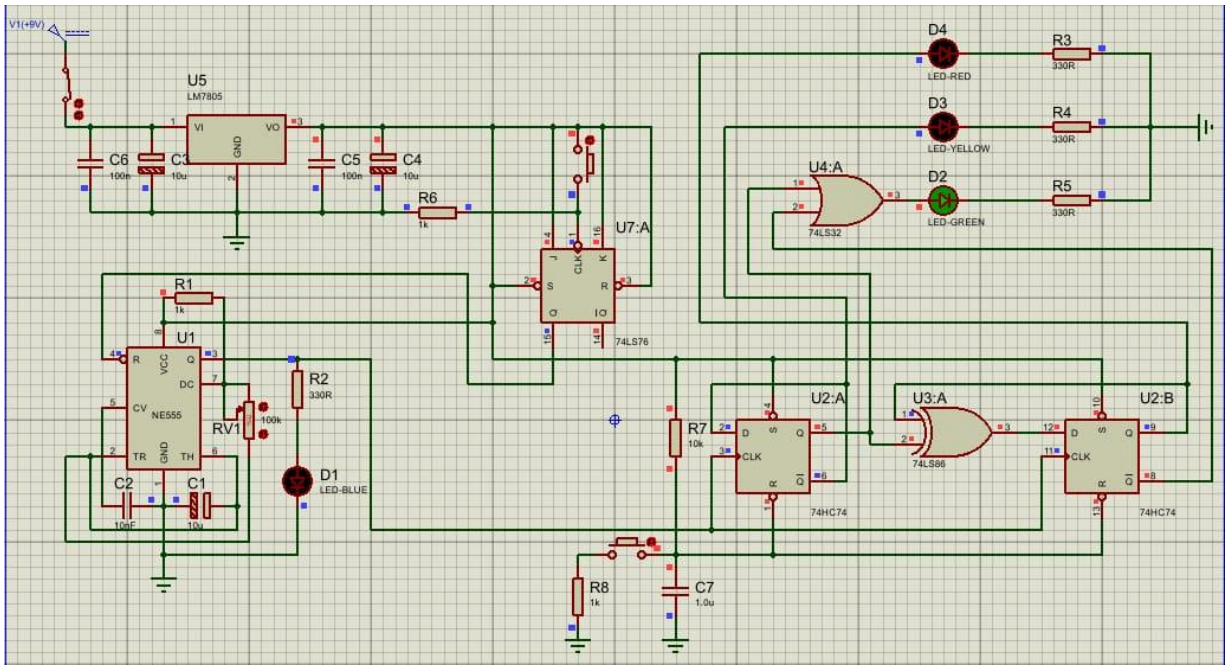
- Breadboard
- Male To Male Jumper
- 9V Battery
- LM7805 Voltage Regulator
- 74LS76AN JK Flip – Flop
- NE555 Timer IC
- 74HC74 D Flip – Flop
- 74LS86 XOR Gate
- 74LS32 OR Gate
- Red LED
- Yellow LED
- Green LED
- Blue LED
- DPDT Switch Button
- 2 × Push Button
- 100k Ω Potentiometer
- 10k Ω Resistor
- 3 × 1k Ω Resistor
- 4 × 330 Ω Resistor
- 3 × 10µF Electrolytic Capacitor (16V)
- 1µF Ceramic Capacitor (16V)
- 2 × 100 nF Ceramic Capacitor (16V)
- 10nF Ceramic Capacitor (16V)

4. Simulated Results:

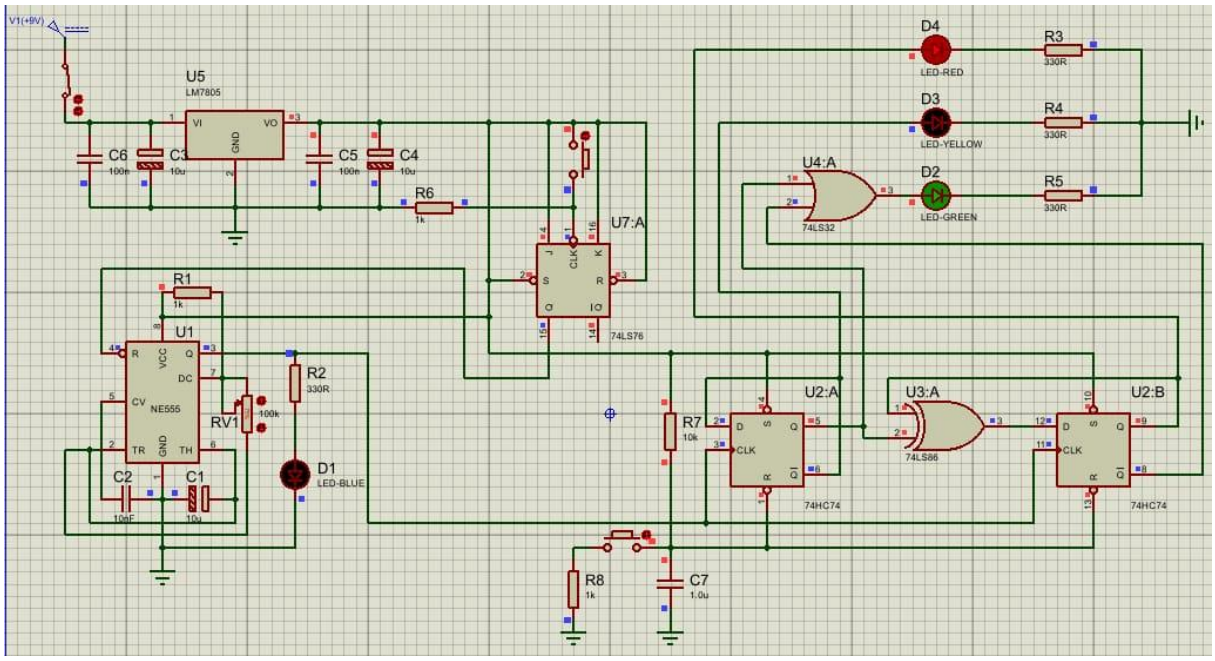
4.1 Starting Condition



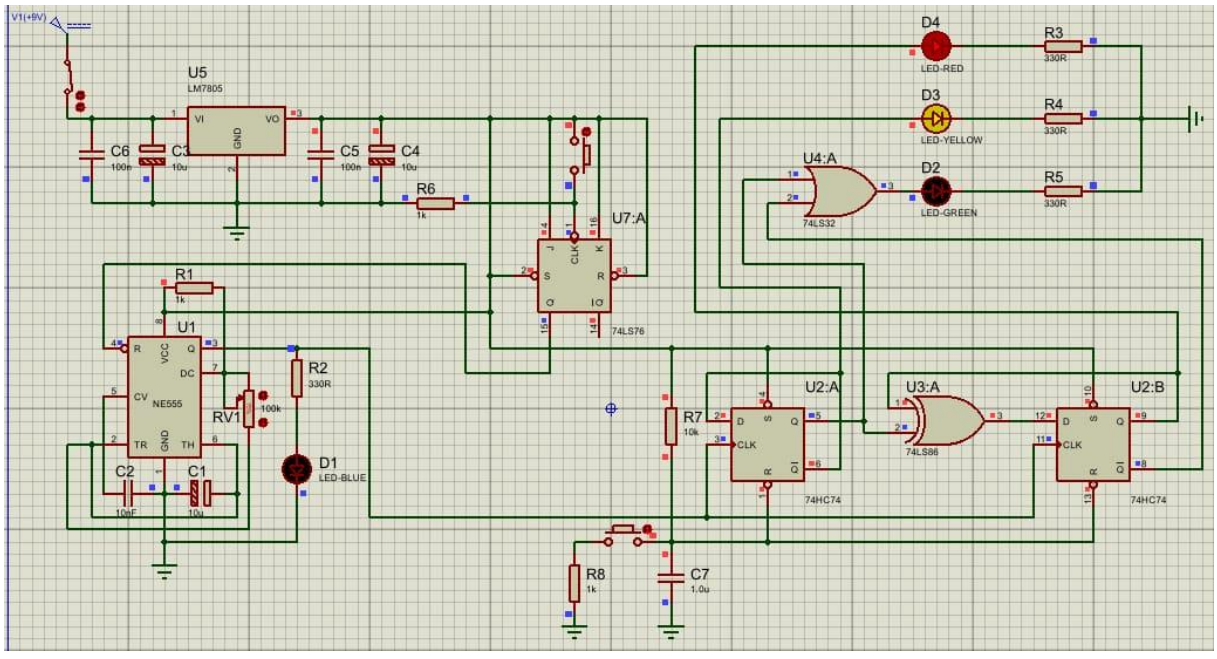
4.2 Second Phase



4.3 Third Phase

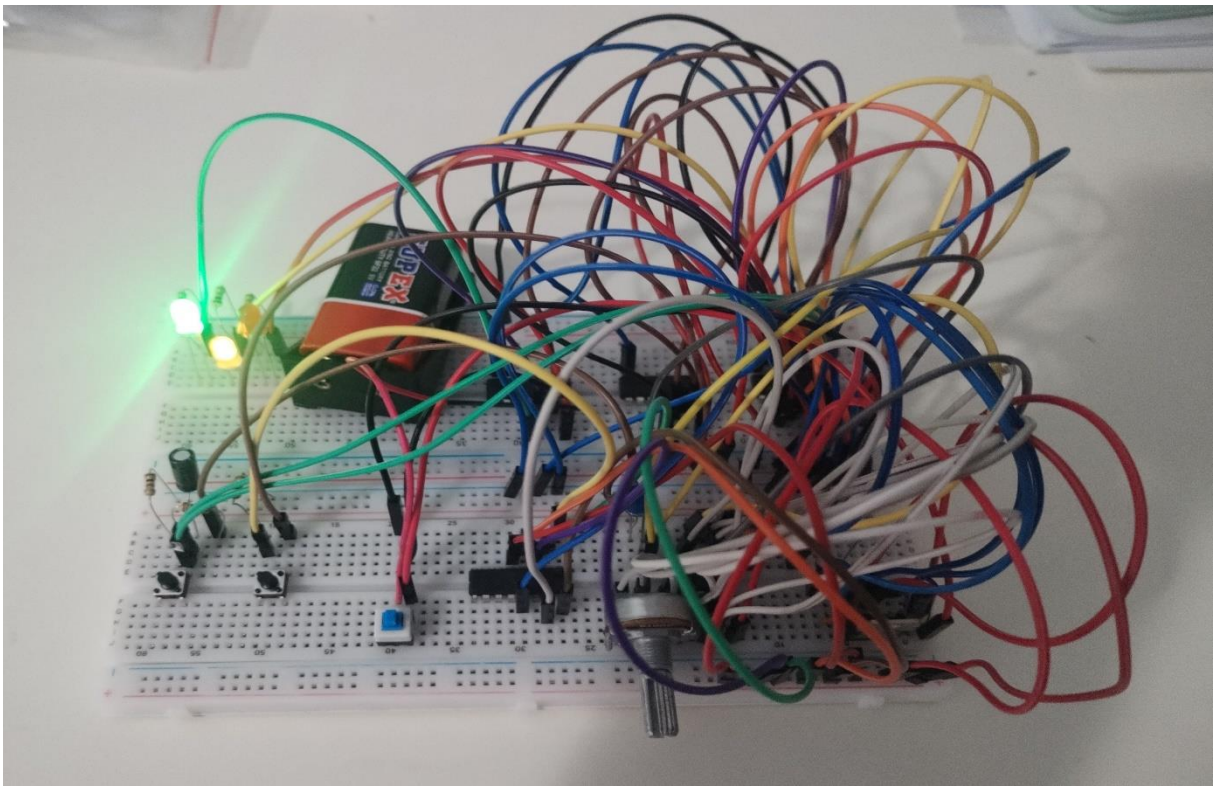


4.4 Fourth Phase

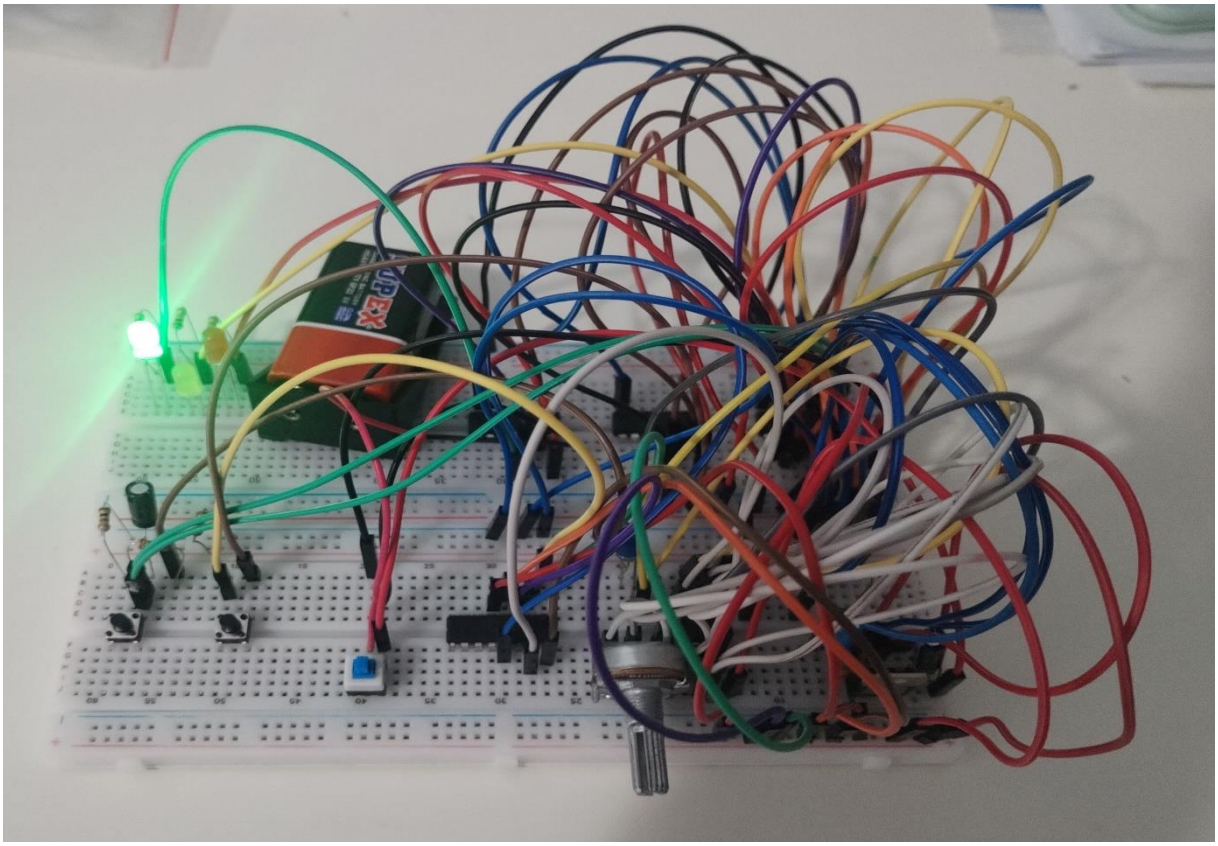


5. Experiment Results:

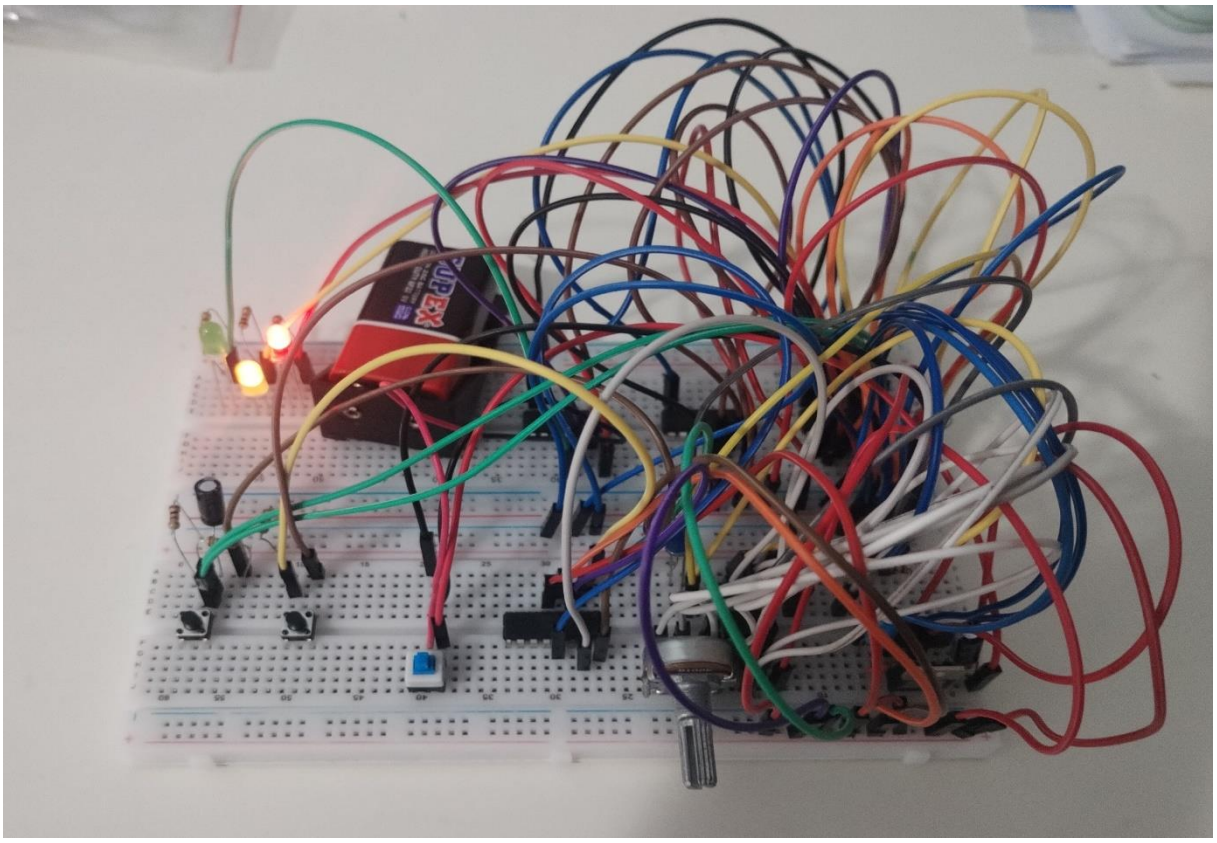
5.1 Starting Condition



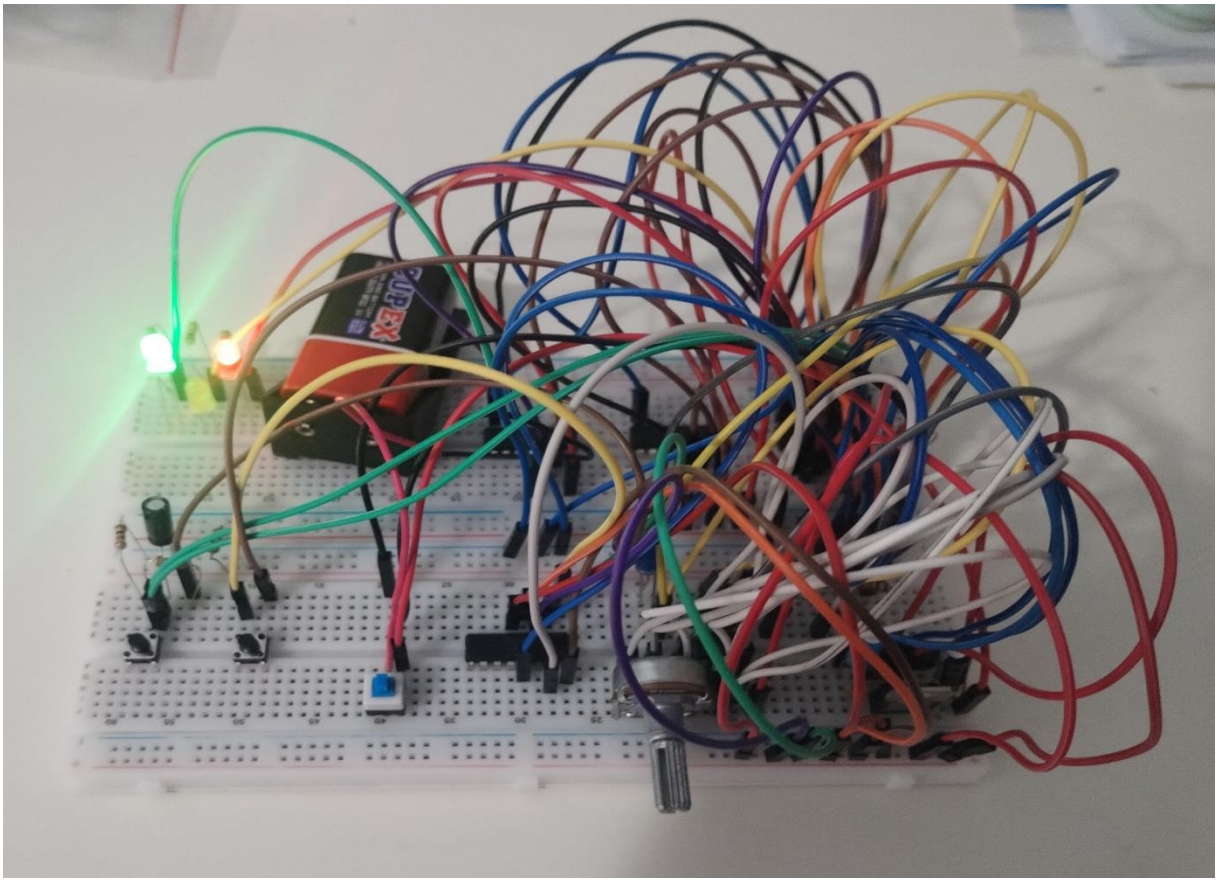
5.2 Second Phase



5.3 Third Phase



5.4 Fourth Phase



6. Conclusions

The completed breadboard circuit successfully demonstrates how fundamental digital logic design principles can be applied to control physical systems. This hardware-based traffic light controller functions as a fully synchronized state machine, utilizing D-type flip-flops and combinational decoding gates to establish a reliable, repetitive signaling sequence. By thoroughly integrating power regulation, astable clock generation, passive RC debouncing, and precise Boolean logic simplification, the project provides profound, hands-on verification of synchronous sequential logic theories without any reliance on software-based microcontrollers.

Personal Engineering Observations:

- **Inverted Protocol:** The implemented $00 \rightarrow 11$ sequence operates as an inverted variation of real-world traffic lights, practically demonstrating the flexibility of custom Boolean synthesis over standard linear cycles.
- **Schmitt Trigger Necessity:** While our RC delay network suffices for this prototype, industrial discrete designs necessitate Schmitt Trigger ICs. These use voltage hysteresis to definitively eliminate mechanical switch chatter and output perfectly sharp digital transitions.
- **MCU Architecture:** If this system were implemented on a modern microcontroller, external debouncing hardware would be entirely obsolete. MCUs natively handle input chatter through internal software algorithms and built-in Schmitt-trigger GPIOs.

7. References

[1] Texas Instruments, *SN74LS76A Dual J-K Flip-Flops With Clear Datasheet*, SDLS121, Rev. 2003. [Online]. Available: https://www.ti.com/lit/ds/symlink/sn74ls76a.pdf

[2] Texas Instruments, *SN74HC74 Dual D-Type Positive-Edge-Triggered Flip-Flops With Clear and Preset Datasheet*, SCLS115, Rev. 2003. [Online]. Available: https://www.ti.com/lit/ds/symlink/sn74hc74.pdf

[3] Texas Instruments, *NE555 Precision Timers Datasheet*, SLFS022, Rev. 2014. [Online]. Available: https://www.ti.com/lit/ds/symlink/ne555.pdf

[4] Texas Instruments, *SN74LS86A Quadruple 2-Input Exclusive-OR Gates Datasheet*, SDLS033, Rev. 2003. [Online]. Available: https://www.ti.com/lit/ds/symlink/sn74ls86a.pdf

[5] Texas Instruments, *SN74LS32 Quadruple 2-Input OR Gates Datasheet*, SDLS033, Rev. 2003. [Online]. Available: https://www.ti.com/lit/ds/symlink/sn74ls32.pdf

[6] ON Semiconductor, *LM7805 1A Positive Voltage Regulator Datasheet*, Rev. 2014. [Online]. Available: https://www.onsemi.com/pdf/datasheet/lm7805-d.pdf

[7] M. M. Mano and M. D. Ciletti, *Digital Design: With an Introduction to the Verilog HDL*, 5th ed. Upper Saddle River, NJ: Pearson, 2013.

[8] J. Ganssle, "A Guide to Debouncing," *The Ganssle Group*, 2008. [Online]. Available: https://my.eng.utah.edu/~cs5780/debouncing.pdf